

Stress Prediction from Wrist Physiological Signals

Using the WESAD Dataset

Machine Learning and Deep Learning for Healthcare — SUPSI MeDiTech/BSP 2026

Student names: Biddiscombe, Daans, Kubišová, Sedra

Link to the repository: <https://github.com/LouieDaans232345/healthcare-project>

1. Introduction

Long-term stress is linked to serious health consequences including cardiovascular disease, impaired immune function, and mental health disorders. Automated, continuous stress monitoring using wearable devices offers a promising approach to early detection and intervention in real-world settings. This project develops machine learning models to classify four affective states — baseline, stress, amusement, and meditation — from physiological signals recorded by a wrist-worn Empatica E4 smartwatch sensor, using the WESAD dataset (Schmidt et al., 2018). We focus exclusively on wrist-based signals, making the approach directly applicable to consumer-grade wearables. This is also a mandatory requirement of the project specification.

The original WESAD benchmark achieved 76% accuracy on a three-class problem (baseline, stress, amusement) using ensemble methods under leave-one-subject-out (LOSO) cross-validation on wrist physiological features. We extend this in two directions: (1) we include meditation as a fourth affective class, motivated by the paper's own conclusion (Schmidt et al., 2018, Section 6); and (2) we evaluate a different set of models — Logistic Regression, Linear SVM, XGBoost, and MLP — providing a complementary comparison to the paper's Random Forest and AdaBoost benchmark. We focus exclusively on the four-class formulation, evaluating both the full physiological feature set (41 features) and a reduced set of 20 features selected via correlation-based clustering and XGBoost importance ranking.

2. Dataset and Data Understanding

2.1 Dataset Description

WESAD is a publicly available multimodal dataset for wearable stress and affect detection (Schmidt et al., 2018). It contains physiological and motion data from 15 subjects (originally 17; S1 and S12 excluded due to sensor malfunction) recorded in a controlled lab setting. Subjects followed one of two counterbalanced protocol versions, both consisting of: a 20-minute neutral baseline, a stress condition using the Trier Social Stress Test (TSST — public speaking and mental arithmetic), an amusement condition (funny video clips), and two guided meditation periods between conditions. The wrist-worn Empatica E4 records four modalities: Blood Volume Pulse (BVP, 64 Hz), Electrodermal Activity (EDA, 4 Hz), Skin Temperature (TEMP, 4 Hz), and 3-axis Accelerometer (ACC, 32 Hz). Labels are stored at 700 Hz (RespiBAN synchronization rate), requiring downsampling before alignment with wrist signals.

2.2 Label System and Class Selection

The dataset contains eight label values (0–7). We retain four conditions: Baseline (label 1, ~19 min), Stress (label 2, ~10 min), Amusement (label 3, ~6 min), and Meditation (label 4, ~13 min). Labels 0, 5, 6, and 7 are discarded as they represent undefined periods and brief transitions between conditions. Meditation is retained as a fourth class despite being excluded from the paper's benchmark, motivated by Schmidt et al. (2018) Section 6, which explicitly suggests it as a future direction. The class distribution is structurally imbalanced across all 15 subjects (baseline ~39%, meditation ~26%, stress ~22%, amusement ~12%), motivating the use of macro F1-score as the primary evaluation metric, consistent with the original paper.

2.3 Inter-Subject Variability

Raw signal analysis across all 15 subjects revealed substantial inter-subject variability. EDA during the stress condition shows the most dramatic differences — mean conductance ranges from 0.34 μS (S14) to 12.84 μS (S13), a nearly 40x difference, driven by individual differences in sweat gland density and sympathetic nervous system reactivity. BVP amplitude (standard deviation) ranges from 18 to 165 across subjects, a 10x difference. TEMP is the most consistent signal (30–34°C range). Condition durations are highly consistent across all subjects (baseline ~19 min, stress ~11–12 min, amusement ~6 min), confirming the class imbalance is a structural property of the protocol and not subject-specific. The two protocol versions (A: baseline→amusement→stress; B: baseline→stress→amusement) were successfully identified from the timelines for all 15 subjects, consistent with Schmidt et al. (2018), Figure 2. This large inter-subject variability confirms that subject-level normalization inside the LOSO loop is essential to prevent models from learning subject identity rather than affective state.

3. Methods

3.1 Signal Preprocessing

All preprocessing is applied to wrist signals only. The following steps are applied to each subject before segmentation:

- BVP: 4th order Butterworth bandpass filter (0.5–8 Hz, zero-phase) to remove baseline wander and high-frequency noise while preserving the cardiac component. Motion artifact spikes within the passband are not fully removed — noted as a limitation.
- EDA: 4th order Butterworth lowpass filter (1 Hz cutoff, zero-phase). The paper specifies 5 Hz, but at a 4 Hz sampling rate the Nyquist frequency is 2 Hz, making a 5 Hz cutoff ineffective. We use 1 Hz to achieve meaningful smoothing while preserving tonic (SCL) and phasic (SCR) components. Documented deviation from the paper.
- TEMP: 5-second moving average to smooth sensor quantization artifacts while preserving slow physiological trends. Minor deviation from the paper, which uses raw TEMP.
- ACC: No filter applied. 3D magnitude computed as $\sqrt{x^2+y^2+z^2}$, consistent with Schmidt et al. (2018) Table 1. A 1-second moving average is applied to the magnitude inside the segmentation loop to reduce high-frequency jitter. Both axes and magnitude are retained for feature extraction.

3.2 Sliding Window Segmentation

Signals are segmented using a 60-second sliding window with a 0.25-second step (99.6% overlap), following Schmidt et al. (2018) Section 4.1. Each window is assigned the label occurring most frequently in the corresponding label array segment (majority vote at 700 Hz). Windows whose majority label is not one of the four conditions of interest are discarded. This produced 179,817 windows across 15 subjects (~12,000 per subject), with class distribution matching the protocol structure: baseline 39.2%, meditation 26.3%, stress 22.2%, amusement 12.4%.

3.3 Feature Extraction

Features are extracted per window using NeuroKit2. The feature set closely follows Schmidt et al. (2018) Table 1, adapted for wrist-only signals (EMG and RESP are chest-only and excluded). The resulting feature matrix contains 179,817 rows and 46 columns (3 identifiers + 41 physiological features + 2 validity flags):

- BVP/HRV (11 features): Heart rate mean/std, HRV time-domain (mean, std, NN50, pNN50, RMSSD) and frequency-domain (ULF, LF, HF, UHF, LF/HF ratio). A validity flag `bvp_hrv_freq_valid` marks windows with insufficient signal quality for frequency analysis (0.008% of windows).
- EDA (9 features): Statistical (mean, std, min, max, slope, range) and phasic SCR features (peak count, mean amplitude, area under curve) via NeuroKit2 decomposition. A flag `eda_scr_has_peaks` identifies windows with no detected peaks (16.5% — physiologically valid for low-arousal states).
- TEMP (6 features): Mean, std, min, max, range, slope.
- ACC (15 features): Per-axis mean/std (x, y, z), 3D magnitude mean/std, per-axis absolute integral, per-axis peak frequency.

No null values were found in the feature matrix — fallback imputation in the extraction pipeline handled all NeuroKit2 failures. The two flag columns are dropped before modelling, leaving 41 physiological features.

3.4 Feature Selection

Feature selection follows a two-step approach. First, features with absolute pairwise correlation ≥ 0.90 are grouped into clusters using hierarchical clustering with average linkage. This produced 33 clusters, correctly grouping the 8 highly correlated pairs identified in the correlation analysis (e.g., `temp_mean/min/max`, `eda_mean/min/max`, `bvp_hrv_std/rmssd`). One representative per cluster is selected based on highest variance. Second, an XGBoost classifier is fit on the full dataset using these 33 representatives, and the top 20 features by importance score are retained as the selected feature set. The three most important features are `acc_y_std` (0.172), `eda_scr_auc` (0.153), and `acc_z_std` (0.113), together accounting for 44% of total importance. ACC features dominate the top 20 (9 of 20), which is noted as a potential concern since ACC magnitude showed little class separation in the feature distribution analysis — this may reflect subject-specific posture patterns rather than genuine affective state differences, and is a consequence of performing feature selection outside the LOSO loop.

3.5 Modelling and Validation

Four models are evaluated under LOSO cross-validation: Logistic Regression (`C=1`, `L2` penalty, `class_weight=balanced`), Linear SVM (wrapped in `CalibratedClassifierCV` for probability estimates, `C=1`, `class_weight=balanced`), XGBoost (100 estimators, max depth 6, learning rate 0.1), and MLP (hidden layers 100 and 50 units, `alpha=0.001`, early stopping). Random Forest was excluded due to computational constraints with

179,817 windows across 15 LOSO folds. All models are evaluated on both feature sets. Inside each fold, StandardScaler is fit on the 14 training subjects only and applied to the held-out test subject, preventing data leakage. The primary evaluation metric is macro F1-score, consistent with Schmidt et al. (2018), Section 4.3.

3.6 Results

Table 1 reports mean macro F1-score and accuracy across 15 LOSO folds for all four models on both feature sets. All models were evaluated on the 4-class problem (baseline, stress, amusement, meditation).

Model	F1 (Full)	Acc (Full)	F1 (Selected)	Acc (Selected)	Best FS
Logistic Regression	0.524 ± 0.101	0.576 ± 0.107	0.506 ± 0.116	0.564 ± 0.109	Full
XGBoost	0.514 ± 0.087	0.625 ± 0.084	0.519 ± 0.091	0.629 ± 0.081	Selected
Linear SVM	0.507 ± 0.087	0.651 ± 0.086	0.487 ± 0.091	0.631 ± 0.091	Full
MLP	0.478 ± 0.142	0.573 ± 0.140	0.434 ± 0.179	0.533 ± 0.173	Full

Table 1: LOSO cross-validation results (mean ± std across 15 subjects). F1 = macro F1-score, Acc = accuracy, FS = feature set.

Logistic Regression achieved the highest macro F1-score overall (0.524 ± 0.101) using all 41 features, while XGBoost achieved the highest accuracy (0.629 ± 0.081) on the selected 20 features. The discrepancy between F1 and accuracy rankings reflects the class imbalance — Linear SVM achieves the highest accuracy (0.651) on all features but a lower F1 (0.507), suggesting it learns to predict majority classes (baseline and meditation) more aggressively at the expense of amusement.

Feature selection did not consistently improve performance: for Logistic Regression and Linear SVM, reducing to 20 features lowered macro F1 by 0.018 and 0.020 respectively. XGBoost showed a slight improvement (+0.005 F1) with selected features. MLP showed the largest degradation (-0.043 F1), likely due to reduced feature diversity limiting the network’s ability to learn complex patterns.

The confusion matrices reveal a consistent pattern across all models: stress is classified best (diagonal values 0.72–0.80), baseline reasonably well (0.56–0.80), while amusement is the hardest class to distinguish (diagonal values 0.00–0.35), frequently confused with baseline and meditation. This is expected given amusement’s short protocol duration (6 min, 12.4% of data) and physiological overlap with other relaxed states.

ROC analysis for the best model (Logistic Regression, all features) confirms that stress is by far the most separable class (AUC=0.94), followed by meditation (AUC=0.84) and baseline (AUC=0.83). Amusement has a near-random ROC (AUC=0.54), confirming it is not reliably distinguishable from wrist physiological signals alone using the current feature set.

Per-subject F1 heatmaps reveal substantial variability in model performance across subjects, ranging from 0.25 to 0.72 depending on subject and model. Subjects S5, S14, and S17 are consistently among the best predicted, while S2, S15, and S16 show consistently lower performance. A scatter plot of per-subject F1 against mean EDA variability during stress shows a weak positive correlation ($r=0.24$), suggesting subjects with stronger EDA responses during stress produce clearer physiological signatures that are easier to classify. However with only 15 subjects this finding should be interpreted cautiously, and the weak correlation indicates other subject-specific factors also drive performance differences.

Feature importance analysis reveals that ACC variability features dominate the selected feature set — acc_y_std (0.172) and acc_z_std (0.113) together account for nearly 30% of total XGBoost importance, with eda_scr_auc (0.153) as the only non-ACC feature in the top 3. This is somewhat counterintuitive given that BVP heart rate showed the clearest visual class separation in the feature distribution analysis, and suggests XGBoost may be leveraging ACC features as proxies for subject identity or posture rather than genuine stress physiology — a risk of performing feature selection outside the LOSO loop.

The full vs selected feature comparison confirms that feature selection did not consistently benefit performance: 3 of 4 models degraded with the top 20 features (Logistic Regression -0.018, Linear SVM -0.020, MLP -0.043), while only XGBoost showed a marginal improvement (+0.005). The MLP degradation is the largest, consistent with neural networks benefiting more from feature diversity. Overall the full 41-feature set is preferred for macro F1, though the selected set achieves higher accuracy for XGBoost and Linear SVM due to reduced overfitting on majority classes.

Compared to the Schmidt et al. (2018) wrist-only benchmark of 76% accuracy (3-class, RF/AdaBoost), our best model achieves 57.6% accuracy on a harder 4-class problem including meditation. The gap is expected given the increased difficulty of the 4-class task and our exclusion of Random Forest due to computational constraints.

4. Critical Analysis and Healthcare Considerations

4.1 Can More Advanced AI Techniques Be Applied?

Yes. The current pipeline uses hand-crafted features from segmented windows fed into classical ML models. More advanced approaches include: (1) end-to-end deep learning on raw signals using 1D CNNs or LSTMs, which could learn task-relevant representations without manual feature engineering; (2) Transformer-based architectures for long-range temporal dependency modelling; (3) multi-task learning to jointly optimize multiple classification objectives; and (4) self-supervised pre-training on unlabelled physiological data to improve generalization with limited labelled subjects.

4.2 Data Quality and Access

The WESAD dataset is publicly available and well-documented. Key challenges include: (1) motion artifacts in BVP are not fully removed by bandpass filtering, as large motion spikes fall within the cardiac passband; (2) wrist EDA at 4 Hz provides limited temporal resolution compared to chest EDA; (3) only 15 subjects are available, leaving 14 for training per LOSO fold; and (4) the lab-controlled setting with student participants may not generalize to real-world stress scenarios with diverse populations.

4.3 Bias, Fairness, and Inclusion

All 15 participants were graduate students at a single research facility, mean age 27.5 years, 12 male and 3 female. This homogeneous population raises concerns about generalizability: physiological stress responses differ by age, sex, and health status; and with only 3 female participants the dataset is insufficiently powered to assess sex-based differences. Any deployed model trained on this data would risk systematic bias against underrepresented groups.

4.4 Explainability (XAI)

Our pipeline supports interpretability at multiple levels: XGBoost feature importance provides signal-level insight into which physiological modalities drive predictions, with ACC variability features (`acc_y_std`, `acc_z_std`) and EDA phasic features (`eda_scr_auc`) dominating. SHAP values could be computed per subject to explain individual predictions. However, the dominance of ACC features in the importance ranking — despite ACC showing poor class separation in the boxplot analysis — highlights a risk of feature importance reflecting spurious correlations with subject identity rather than genuine stress physiology.

4.5 Ethics and GDPR

Stress detection from wearable data constitutes processing of sensitive health data under GDPR Article 9, requiring explicit informed consent and data minimization. Continuous workplace stress monitoring could be misused by employers. Any real-world deployment would require a full ethical review, data protection impact assessment, and transparent communication to users about what is being monitored and why.

4.6 Generalizability

LOSO cross-validation provides subject-independent evaluation, the standard approach for wearable stress datasets (Schmidt et al., 2018, Section 4.3). However, external validation on an independent dataset has not been performed and would be needed before any clinical deployment. The large inter-subject variability — particularly the 40x EDA range — suggests generalizability to truly novel populations remains a significant open challenge.

5. Limitations and Future Work

The main limitations of this work are: fixed hyperparameters chosen from literature rather than tuned inside the LOSO loop; feature selection performed outside LOSO on the full dataset, introducing a potential leakage risk; no artifact removal for BVP motion spikes within the cardiac passband; exclusion of Random Forest and RBF SVM due to computational constraints; a small dataset of 15 subjects limiting statistical power; and a lab-controlled protocol that may not generalize to real-world stress scenarios. The addition of meditation as a fourth class, while scientifically motivated, significantly increases classification difficulty — particularly since amusement and meditation share similar physiological signatures at the wrist. Future work should address these through LOSO-internal hyperparameter tuning, end-to-end deep learning on raw signals, and external validation on independent datasets.

References

- Schmidt, P., Reiss, A., Duerichen, R., Marberger, C., & Van Laerhoven, K. (2018). Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection. In Proceedings of the 20th ACM International Conference on Multimodal Interaction (ICMI 2018), 391–399.
- Kreibig, S. D. (2010). Autonomic nervous system activity in emotion: A review. *Biological Psychology*, 84(3), 394–421.
- Makowski, D., et al. (2021). NeuroKit2: A Python toolbox for neurophysiological signal processing. *Behavior Research Methods*, 53(4), 1689–1696.